

APS110 – Lecture 33

See hand written notes for drawings.

Thermodynamics

- Under the right conditions (exposed to the atmosphere and at a constant temp) we can massage our equations to get Gibbs free energy
 - $\Delta G = \Delta H - T\Delta S$
 - Spontaneous when $\Delta G < 0$
 - NOTE: this satisfies the requirement that entropy of the universe is increasing
- Types of Rxns
 - Combustion of a hydrocarbon → **should be spontaneous at ANY temperature**
 - $\Delta H = -ve$ (exothermic)
 - $\Delta S = +ve$
 - Freezing of water
 - $\Delta H = -ve$ (exothermic)
 - $\Delta S = -ve$ (frozen water is more ordered than liquid)
- **Metastable** → a metastable sample of water could be brought below freezing temperature and remain in liquid state if it is undisturbed
 - If you were to disturb this sample, it would create a point for enucleation to initiate the second phase → solid front will progress through the sample
 - The water will then be a slush, because as it changes phase it is an exothermic rxn – water was at negative temp, but as it freezes it releases heat so slush forms
- Consider water
 - Plot $G = H - TS \rightarrow G = -TS + H$ ($y = mx + b$)
 - Plot Gibbs energy vs. temp
 - Slope is the entropy → negative slope
 - Phases:
 - Gas → has highest entropy, thus steep slope
 - Liquid → less entropy, less slope
 - Where these two lines intersect is going to be a temperature beyond which the gas phase will be STABLE
 - Vapour phase will be favoured above this temp
 - Solid → lowest entropy
 - Solid and liquid intersection gives temperature below which the solid phase is stable
- Consider CO_2
 - The line for gas phase plunges down (rapid change in entropy) so that it intersects the solid line before liquid line does
 - Above this temperature the solid will change directly into gas phase (sublimation)
 - Think of dry ice
- Pressure will change the equilibrium conditions as well
 - Many things change equilibrium conditions → ex. chemical composition
 - Plot pressure vs. temperature (CO_2)

- CO_2 has covalent bonding → we would expect high bond energy
 - Also has high band gap
 - Because of this high band gap, we would expect it to be clear (photons pass right through)
 - We should look for this optical change when solid CO_2 (dry ice) changes to liquid CO_2
- We can encourage different phase changes by manipulating temperature and pressure
- DEMO: Dry ice
 - Ramsay wants to encourage the sublimation of dry ice to liberate as much gas as possible → will use hot water
 - As it sublimates, it liberates vapour
 - If we can trap all this vapour in a vessel (ex. bottle), the pressure will go up
 - **Triple point** of CO_2 is at **5 atm**
 - At a certain point, the pressure will be high enough that the solid CO_2 will melt into liquid CO_2
 - There are all three phases in the bottle → solid CO_2 is floating in boiling liquid

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